

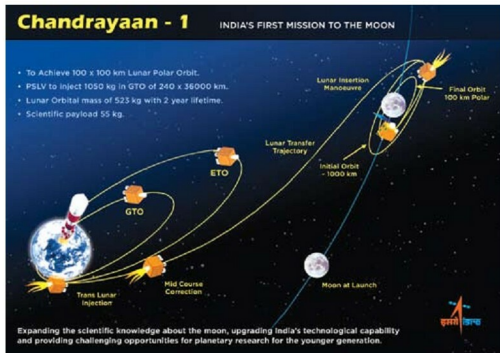


Fig. 2: The mission at a glance

as to a number of asteroids and comets.

A space probe destined for a planet or other astronomical body can be classified as a 'flyby,' 'impactor,' 'orbiter' or 'lander' mission. Historically, flyby missions proved easiest to accomplish, as these did not require the precise navigation needed for an impact, nor the additional propulsion to conduct a manoeuvre to enter the orbit. Upon landing, some landers have released 'rovers,' which travel across the surface of the astronomical body upon which they have landed.

Once a probe has left the vicinity of the earth, its trajectory is likely to take it along an orbit around the sun similar to the earth's orbit. To reach another planet, the conceptually simplest means is to execute a Hohmann transfer orbit manoeuvre.

More complex techniques, such as gravitational slingshots, can be more efficient, though these may require the probe to spend more time in transit. A technique using very little propulsion, but possibly requiring a considerable amount of time, is to follow a trajec-

tory on the interplanetary transport network.

Launch vehicle journey

The polar satellite launch vehicle, usually known by its abbreviation 'PSLV,' is an expendable launch system operated by the Indian Space Research Organisation (ISRO). It was developed to allow India to launch its Indian remote sensing (IRS) satellites into sun synchronous orbits—a service that was, until the advent of the PSLV, commercially viable only from Russia. PSLV can also launch small-size satel-

lites into the geostationary transfer orbit. The reliability and versatility of the PSLV is proven by the fact that it has launched 30 spacecrafts (14 Indian and 16 from other countries) into a variety of orbits so far. In April 2008, it successfully launched ten satellites in one go, breaking the world record held by Russia.

The first Indian moon mission was proposed to be a lunar polar orbiter at an altitude of about 100 km from the lunar surface. Considering the maturity of PSLV demonstrated through PSLV-C4/KALPANA-1 mission, PSLV was chosen for the first lunar mission.

The upgraded version of PSLV, viz, PSLV-C11, which has a liftoff weight of 316 tonnes and is 44.4m tall, was used to inject a 1304kg mass spacecraft at 240x24,000km orbit. The corresponding spacecraft mass was 590 kg when the target lunar orbit of 100 km was achieved.

The PSLVC 11, also called PSLV-XL because of the increased weight of the six strap-on motors, soared into the sky from the second launch pad at the Satish Dhawan Space Centre, Sriharikota. It traveled to the vicinity of the moon by following the lunar transfer trajectory (LTT).

Chandrayaan beams back 40,000 images in 75 days

Chandrayaan-1 has transmitted more than 40,000 images of different types since its launch on October 22, 2008, which many in ISRO believe is quite a record compared to the lunar flights of other nations. ISRO officials estimated that if more than 40,000 images have been transmitted by Chandrayaan's cameras in 75 days, it worked out to nearly 535 images being sent daily. They are first transmitted to Indian Deep Space Network at Bhalalu near Bengaluru, from where they are flashed to ISRO's telemetry, tracking and command network at Bengaluru.

Some of these images have a resolution of up to five metres providing a sharp and clear picture of the moon's surface. In comparison, many images sent by some of the other missions had a 100-metre resolution.

On November 26, the indigenous terrain mapping camera, which was first activated on October 29, 2008, took shots of peaks along with craters. This came as a surprise to ISRO officials because the moon consists mostly of craters.